# Probability Distributions

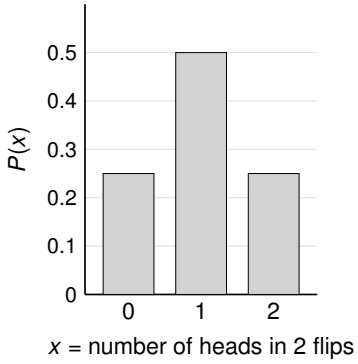
Name _____

Flip a coin 2 times. Let $X$ = the number of heads. $X$ is a **random variable**: a number that comes from a chance process.

Outcomes: HH, HT, TH, TT (all equally likely)

| $x$ (heads) | 0 | 1 | 2 |
|---|---|---|---|
| $P(x)$ | $\frac{1}{4}$ | $\frac{2}{4} = \frac{1}{2}$ | $\frac{1}{4}$ |

This table is the **probability distribution** of $X$. It lists every value of $X$ and its probability.



$x$ = number of heads in 2 flips

**A valid probability distribution:**
1. Every probability is from 0 to 1. 2. The probabilities add to 1: $\Sigma P(x) = 1$

Check: $\frac{1}{4} + \frac{1}{2} + \frac{1}{4} = 1$. Valid.

**Using a distribution.** Add the probabilities of the values you want.
$P(X \geq 1) = P(1) + P(2) = \frac{1}{2} + \frac{1}{4} = \frac{3}{4}$ ($\geq$ means "greater than or equal to")

**Theoretical or empirical?**

| | |
|---|---|
| **Theoretical** | From reasoning about equally likely outcomes, like the coin flips above. |
| **Empirical** | From data. Divide each count by the total.<br>50 families: 0 cars: 5, 1 car: 20, 2 cars: 18, 3 cars: 7<br>$P(0) = \frac{5}{50} = 0.10$, $P(1) = 0.40$, $P(2) = 0.36$, $P(3) = 0.14$ |

Work with a partner. Compare answers when you finish each problem.

1. Is each table a valid probability distribution? Explain.

(a)

| $x$ | 1 | 2 | 3 |
|---|---|---|---|
| $P(x)$ | 0.2 | 0.3 | 0.5 |

(b)

| $x$ | 1 | 2 | 3 |
|---|---|---|---|
| $P(x)$ | 0.4 | 0.4 | 0.3 |

(c)

| $x$ | 1 | 2 | 3 |
|---|---|---|---|
| $P(x)$ | 0.5 | $-0.1$ | 0.6 |

(d)

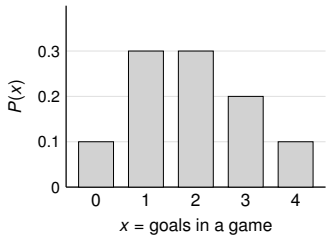
| $x$ | 1 | 2 | 3 |
|---|---|---|---|
| $P(x)$ | $\frac{1}{3}$ | $\frac{1}{3}$ | $\frac{1}{3}$ |

2. Find the missing probability.

| $x$ | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| $P(x)$ | 0.15 | 0.35 | ? | 0.10 |

3. This is the distribution of the number of goals a soccer team scores in a game.

| $x$ | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| $P(x)$ | 0.1 | 0.3 | 0.3 | 0.2 | 0.1 |




(a) $P(X \geq 2)$

(b) $P(X < 1)$

(c) $P(1 \leq X \leq 3)$

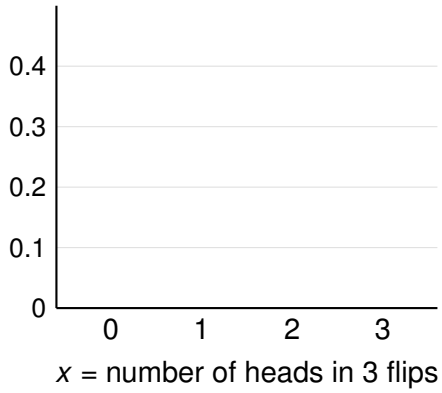
4. Flip a coin 3 times. Let $X$ = the number of heads.

   (a) List all 8 outcomes.

   (b) Make the probability distribution of $X$.

| $x$ | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| $P(x)$ | | | | |

   (c) Draw a bar graph of the distribution.




   (d) Find $P(X \geq 2)$.

5. 40 households said how many pets they have: 0 pets: 12, 1 pet: 16, 2 pets: 8, 3 pets: 4.

   (a) Make the empirical probability distribution.

   (b) Find $P$(at least 1 pet).